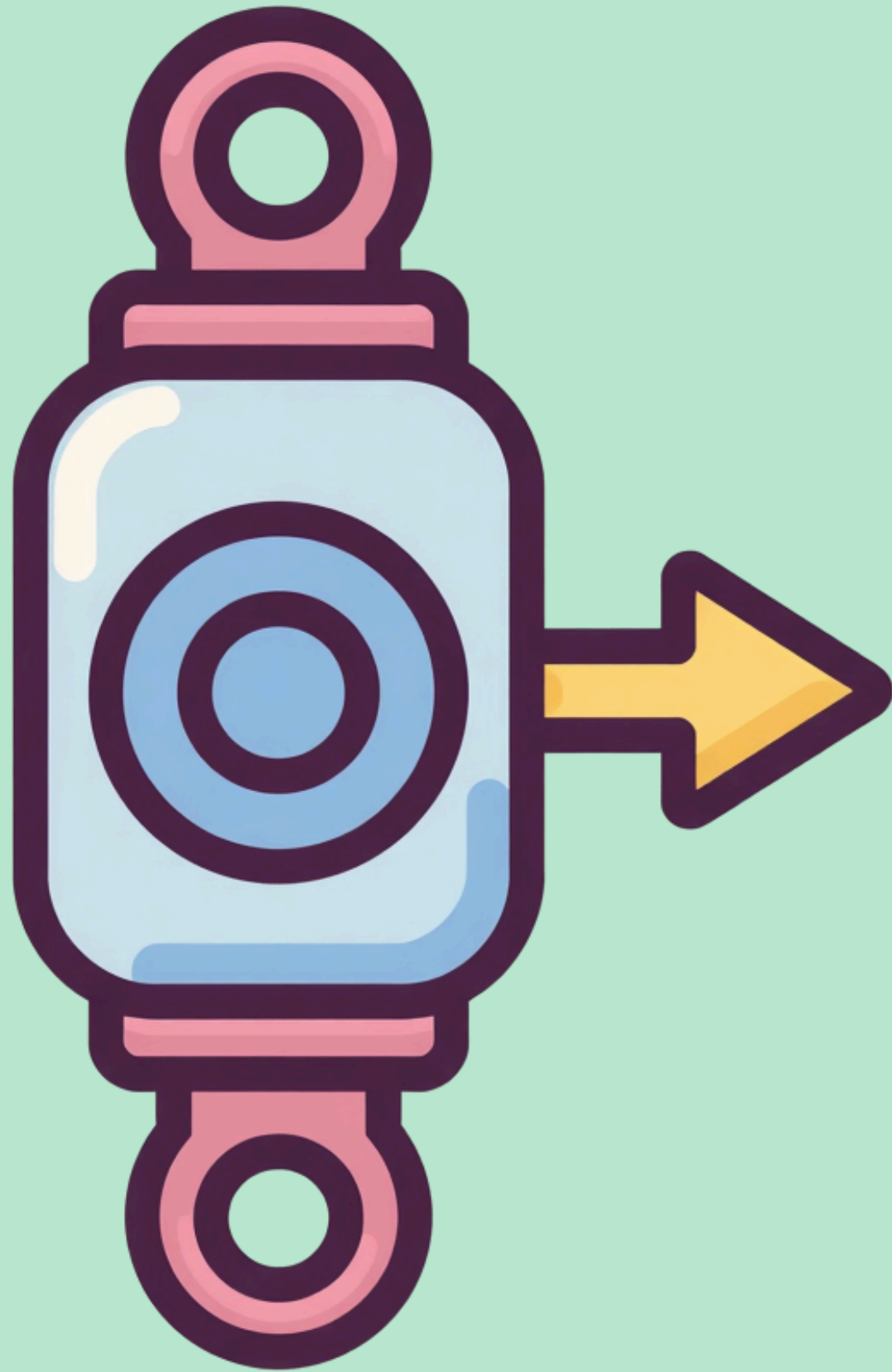
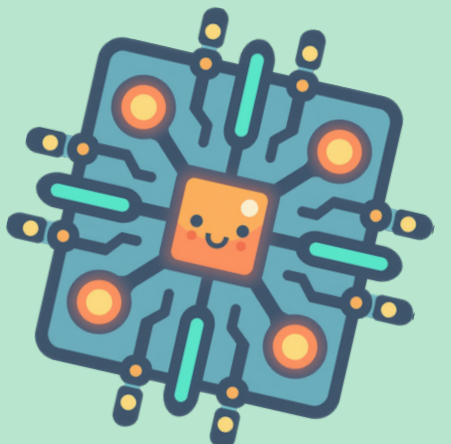
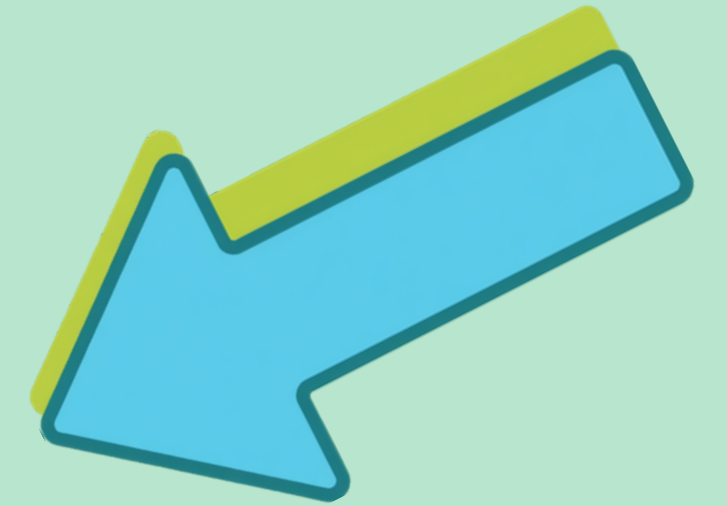




**DIODE – THE
ONE-WAY VALVE
FOR CURRENT**

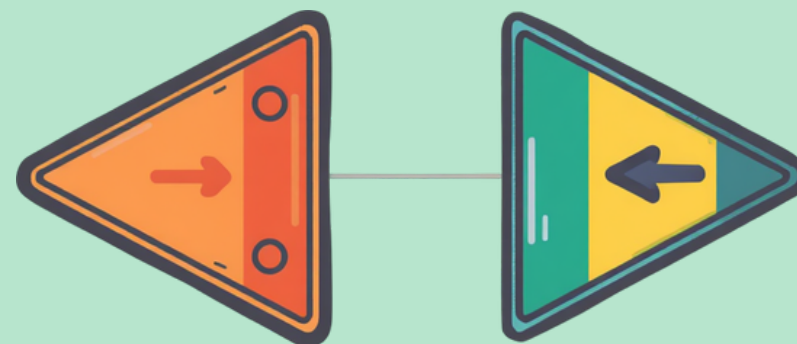
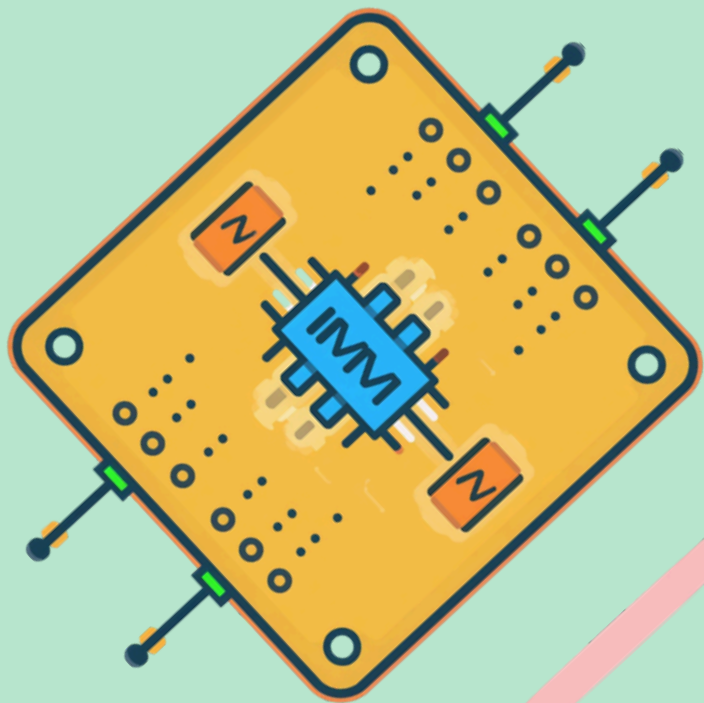




WHAT IS A DIODE?



A diode is a special electronic component that lets electricity flow in only one direction – like a one-way gate! Think of it as an electronic door that only opens in one way. When electricity tries to go the wrong way, the diode blocks it. When it goes the right way, electricity flows through easily. This one-way behavior makes diodes super useful in many devices around us!





DIODE AS A WATER VALVE



- Imagine electricity is like water flowing through pipes! A diode is like a special one-way valve that lets water flow only in one direction. Just like a check valve in plumbing!
- When you push water the right way, it flows through smoothly. But if you try to push it the wrong way, the valve closes and blocks it! A diode works exactly the same way with electricity.
- Think of it like a turnstile at a subway station – you can only pass through in one direction! Electricity can only flow through a diode in one direction too. This keeps it safe and controlled!





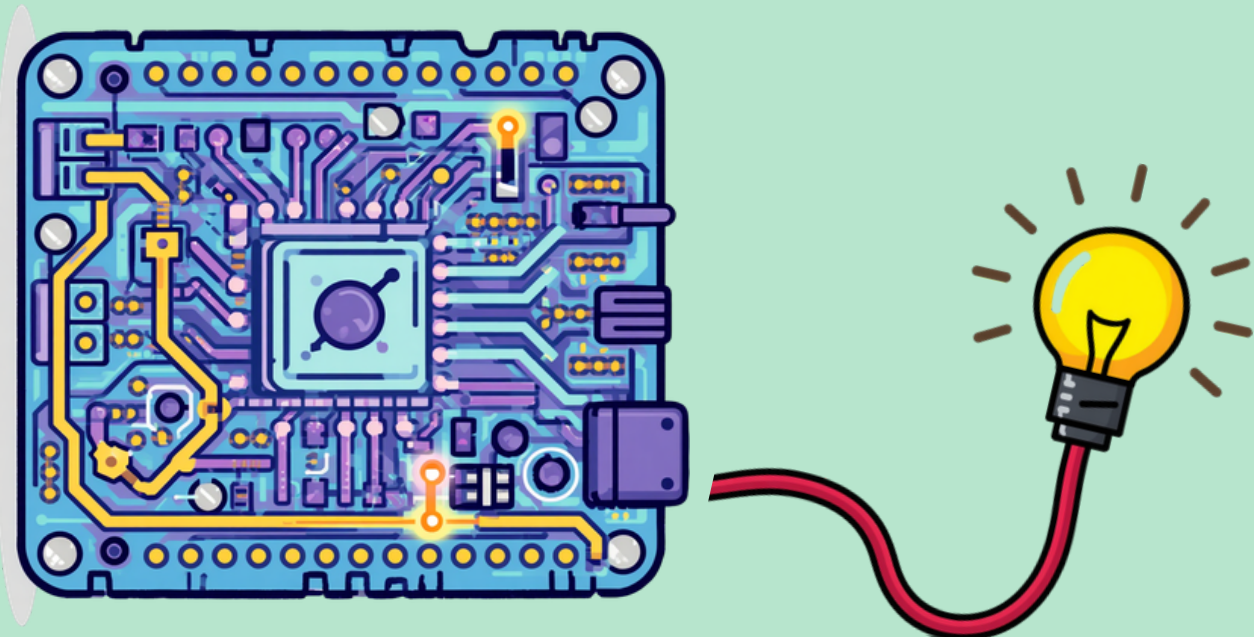
LED EXPERIMENTS



- LEDs are special diodes that light up! They glow when electricity flows through them in the right direction.



- Try swapping the wires – only one way makes the LED glow! This shows how diodes let electricity flow in just one direction.



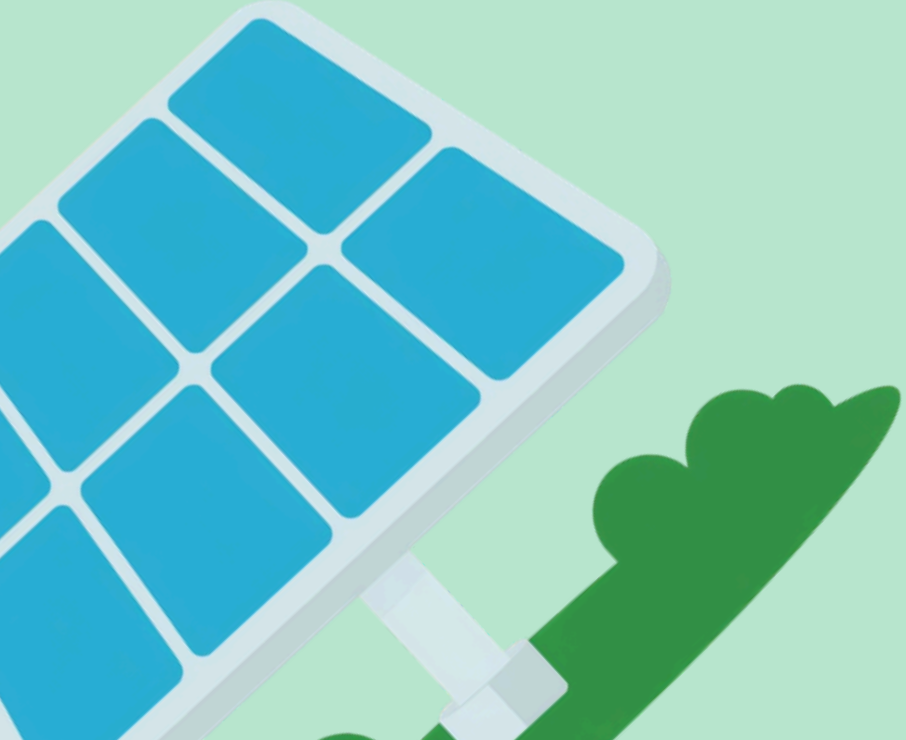


DIODES IN OUR DAILY LIFE



Diwali Lights

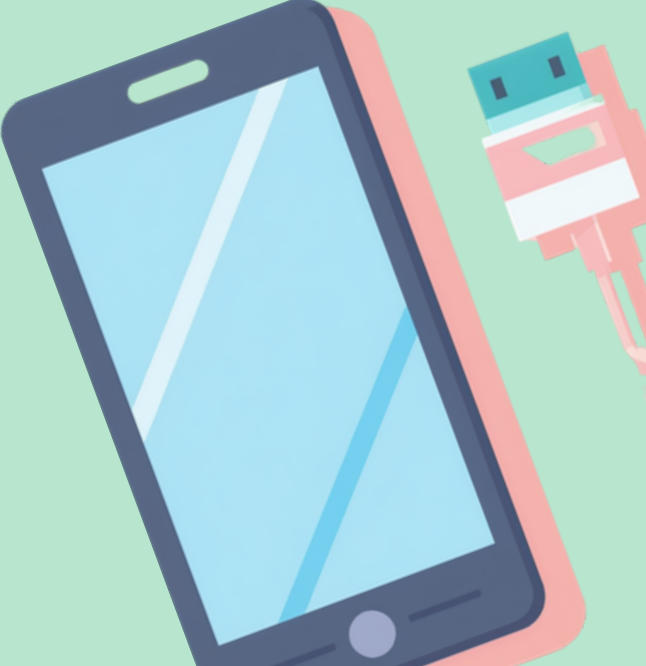
Solar Panels



Mobile Phones

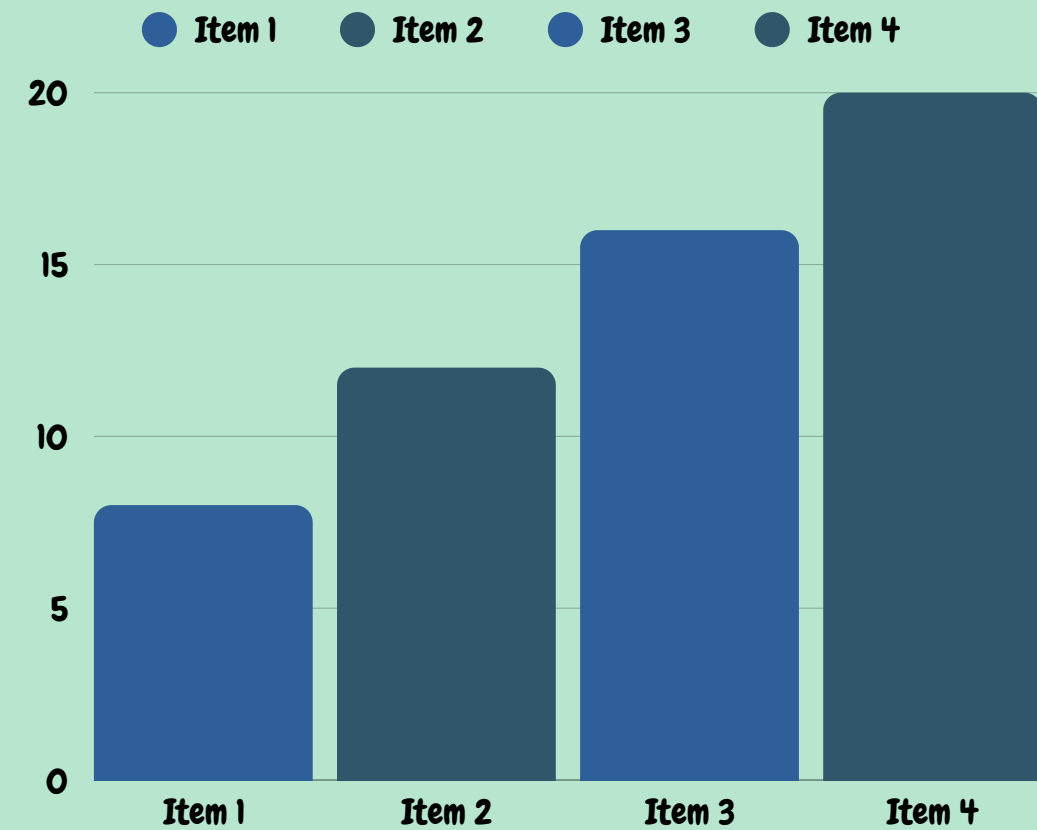
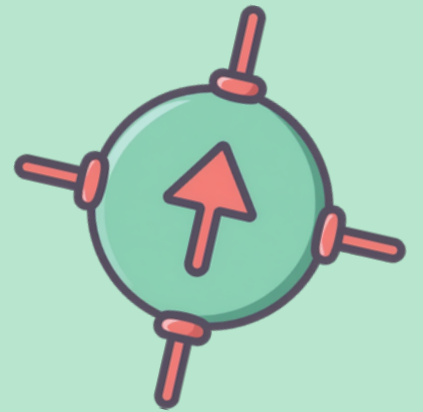
LED Bulbs

TV Remotes

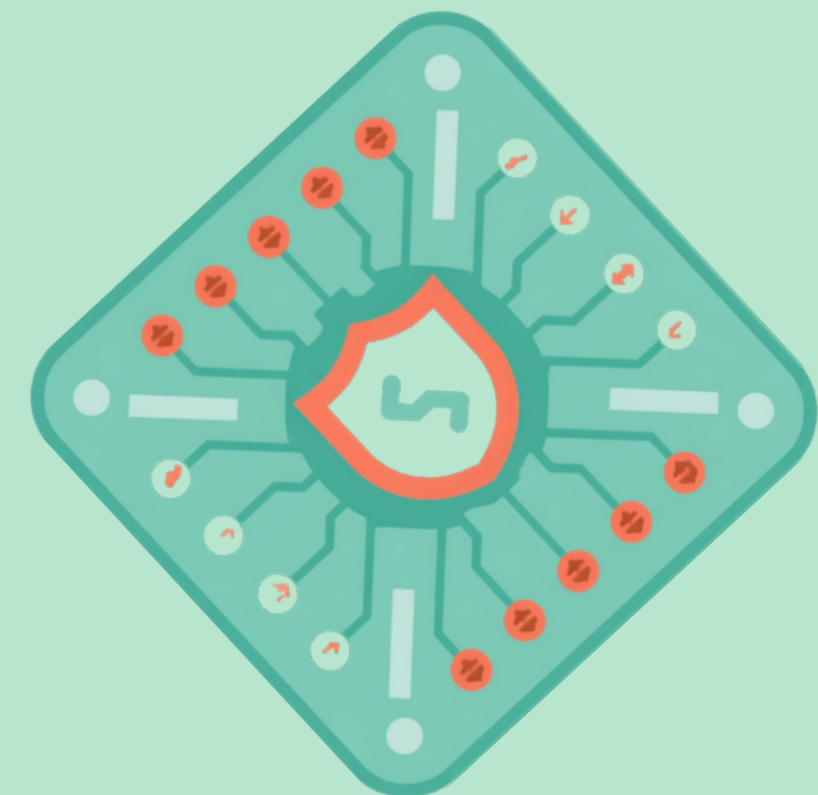


Car Batteries

WHY ARE DIODES IMPORTANT?



- Control electricity flow – only one way! Diodes act like a one-way gate for electric current.
- Protect devices from damage – they prevent dangerous reverse current from harming your gadgets.
- Help save energy with LEDs and solar power – diodes make our lights brighter and solar panels work better!





THANK YOU!

